

# “AI-Powered Humanoid Robot Fleet Management System”

Team: Robo-Nexus

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## Team Members:

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## GitHub Repository:

<https://github.com/sfsu-joseo/csc675-775-database-api-project-robo-nexus>

| Version | Description        |
|---------|--------------------|
| V1      | Initial Submission |
|         |                    |

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## Normalization Table

| Entity             | Failed NF | Reason                                                                                                                                                                         | Fixes                                                                                         |
|--------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Robot              | 3NF       | Manufacturer depends on robot model id not on robot id, and other ids were creating circular dependency.                                                                       | remove manufacturer_id, Performance_metric_id, Charging_station_id,Alert_id from robot table. |
| DeploymentLocation | 3NF       | Technician_id implies a single one for any deployment location, charging session_id creates circular dependency because chargingSession already references DeploymentLocation. | Remove technician_id ,Charging_Session_id                                                     |
| Technician         | 3NF       | No deployment_location_id means that it fails functional requirements                                                                                                          | Add DeploymentLocation_id                                                                     |
| Technician         | 1NF       | Name is not an atomic attribute.                                                                                                                                               | Create a different table with names.                                                          |
| SupportRequest     | 3NF       | User_ID is not dependent on any other attribute and is not prime                                                                                                               | Removed User_ID                                                                               |

|                                     |     |                        |                                                |
|-------------------------------------|-----|------------------------|------------------------------------------------|
| User_Deployme<br>nt_Location        | 1NF | Missing<br>Primary Key | Added<br>UserDeploymen<br>tLocationID          |
| Emotional_Ana<br>lyze               | 1NF | Missing<br>Primary Key | Added<br>EmotionalAnal<br>yzeID                |
| PerformanceDi<br>agnostic           | 1NF | Missing<br>Primary Key | Added<br>PerformanceDi<br>agnosticID           |
| EmotionalInte<br>ractionSessio<br>n | 1NF | Missing<br>Primary Key | Added<br>EmotionalInte<br>ractionSessio<br>nID |
| Technician_Su<br>pport_Request      | 1NF | Missing<br>Primary Key | Added<br>TechnicianSup<br>portRequestID        |
| RobotCharging<br>Station            | 1NF | Missing<br>Primary Key | Added<br>RobotCharging<br>StationID            |
| logSession                          | 1NF | Missing<br>Primary Key | Added<br>logSessionID                          |
| HardwareMaint<br>enance             | 1NF | Missing<br>Primary Key | Added<br>HardwareMaint<br>enanceID             |
| ChargingStati<br>onSession          | 1NF | Missing<br>Primary Key | Added<br>ChargingStati<br>onSessionID          |
| PatientEmotio<br>nalRecord          | 1NF | Missing<br>Primary Key | Added<br>PatientEmotio<br>nalRecordID          |

## List of Main Data Items and Entities (Normalize)

### 1. **Robot:** Strong

- 1.1 robot\_id: key, numeric (PK)
- 1.2 ~~charging\_station\_id (FK)~~ (removed)
- 1.3 robotmodel\_id: numeric (FK)
- 1.4 deployment\_location\_id: numeric (FK)
- 1.5 ~~alert\_id: numeric (FK)~~ (removed)
- 1.6 user\_id: numeric (FK, optional)
- 1.7 software\_version\_id: numeric (FK)
- 1.8 ~~manufacturer\_id: numeric (FK)~~ (removed)
- 1.9 ~~performance\_metric\_id: numeric (FK)~~ (removed)
- 1.10 robotrole\_id: numeric(FK)

### 2. **RobotModel:** Strong

- 2.1 robotmodel\_id: key, numeric (PK)
- 2.2 manufacturer\_id: numeric (FK)
- 2.3 model: enum, alphanumeric

### 3. **User:** Strong

- 3.1. user\_id: key, numeric (PK)
- 3.2. user\_name: alphanumeric

4. **Manufacturer:** Strong
  - 4.1 manufacturer\_id: key, numeric (PK)
  - 4.2 name: enum, alphanumeric
5. **Hardware:** Strong
  - 5.1 hardware\_id: key, numeric (PK)
  - 5.2 manufacturer\_id: numeric (FK)
6. **RobotHardware:** Weak
  - 6.1 robot\_hardware\_id: key, numeric
  - 6.2 robot\_id: numeric (FK)
7. **SoftwareVersion:** Strong
  - 7.1 software\_version\_id: key, numeric
8. **DeploymentLocation:** Strong (Superclass)
  - 8.1 deployment\_location\_id: key, numeric (PK)
  - 8.2 ~~technician\_id: numeric (FK)~~
  - 8.3 address\_id: numeric (FK)
  - 8.4 environment\_type: enum, alphanumeric  
(warehouse/hospital/home/factory)
9. **Address:** Strong
  - 9.1 address\_id: key, numeric (PK)
  - 9.2 street: alphanumeric
  - 9.3 city: alphanumeric
  - 9.4 state: alphanumeric
  - 9.5 zipcode: alphanumeric
10. **Task:** Strong
  - 10.1 task\_id: key, numeric (PK)
  - ~~10.1 ai\_response\_id: numeric (FK)~~
  - 10.2 task\_name: alphanumeric
  - 10.3 description: text
  - 10.4 priority: enum, numeric
  - 10.5 status: bool (Complete)

- 11. **TaskAssignment:** Weak
  - 11.1 task\_assignment\_id: key, numeric (PK)
  - 11.2 robot\_id: numeric (FK)
  - 11.3 user\_id: numeric (FK)
  - 11.4 task\_id: numeric (FK)
  - 11.5 assigned\_at: timestamp
- 12. **TaskExecution:** Weak
  - 12.1 task\_execution\_id: key, numeric (PK)
  - 12.3 task\_id: numeric (FK)
  - 12.4 start\_time: timestamp
  - 12.5 end\_time: timestamp
  - 12.6 status: bool, (Successful)
- 13. **EmotionalRecord:** Weak
  - 13.1 emotional\_id: key, numeric (PK)
  - 13.2 pattern\_id: numeric (FK)
  - 13.3 emotional\_state: enum, alphanumeric
  - 13.4 timestamp: timestamp
  - 13.5 description: text
- 14. **InteractionSession:** Weak
  - 14.1 interaction\_id: key, numeric (PK)
  - 14.2 robot\_id: numeric (FK)
  - 14.3 user\_id: numeric (FK)
  - 14.4 home\_id: numeric (FK)
  - 14.4 start\_time: timestamp
  - 14.5 end\_time: timestamp
- 15. **RobotRole:** Strong
  - 15.1 robotrole\_id: key, numeric (PK)
  - 15.2 user\_id: numeric(FK)
  - 15.3 robotrole: enum, alphanumeric
- 16. **EnergyStatus:** Weak
  - 16.1 energy\_status\_id: key, numeric (PK)

- 16.2 robot\_id: numeric (FK)
- 16.3 charging\_session\_id: numeric (FK)
- 16.4 battery\_level: numeric
- 16.5 timestamp: timestamp
- 17. **PerformanceMetric:** Weak
  - 17.1 performance\_metric\_id: key, numeric (PK)
  - 17.2 robot\_id: numeric (FK)
  - 17.3 metric\_type: alphanumeric
  - 17.4 metric\_value: numeric
  - 17.5 timestamp: timestamp
- 18. **Alert:** Weak
  - 18.1 alert\_id: key, numeric (PK)
  - 18.2 robot\_id: numeric (FK)
  - 18.3 severity: enum, alphanumeric
  - 18.4 timestamp: timestamp
- 19. **DiagnosticReport:** Weak
  - 19.1 diagnostic\_report\_id: key, numeric (PK)
  - 19.2 robot\_id: numeric (FK)
  - 19.3 ai\_model\_id: numeric (FK)
  - 19.4 generated\_at: timestamp
  - 19.5 summary: text
- 20. **Log:** Weak
  - 20.1 log\_id: key, numeric (PK)
  - 20.2 robot\_id: numeric (FK)
  - 20.3 ~~deployment\_location\_id: numeric (FK, optional)~~
  - 20.4 CreatedAt: timestamp
  - 20.5 event\_type: alphanumeric
  - 20.6 event\_details: text
- 21. **Sensor:** Weak
  - 21.1 sensor\_id: key, numeric (PK)
  - 21.2 robot\_id: numeric (FK)

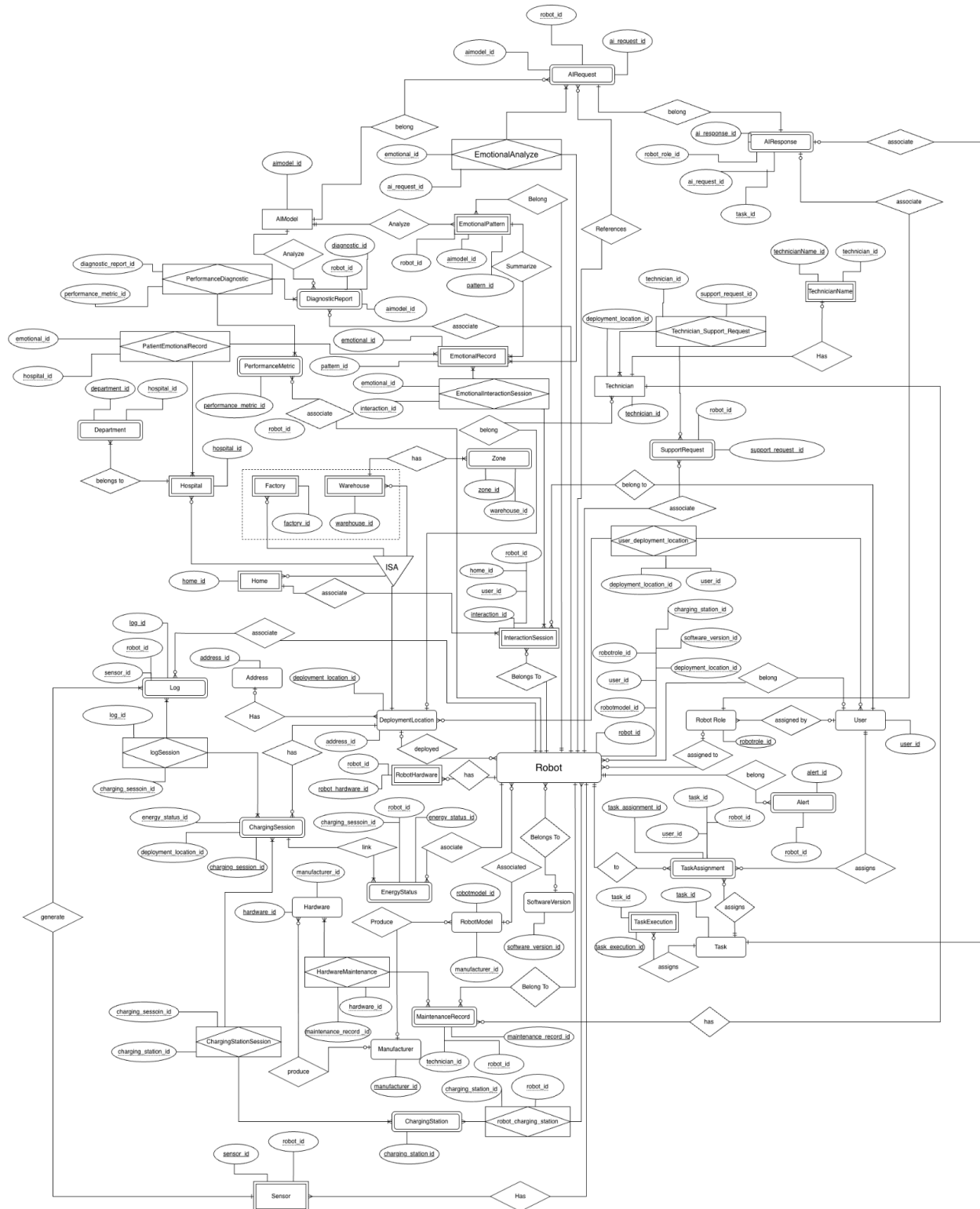
- 21.3 sensor\_type: alphanumeric
- 21.4 status: enum, alphanumeric
- 22. **ChargingStation:** Strong
  - 22.1 charging\_station\_id: key, numeric (PK)
  - 22.2 station\_name: alphanumeric
  - 22.3 capacity: numeric
  - 22.4 status: enum, alphanumeric
- 23. **ChargingSession:** Weak
  - 23.1 charging\_session\_id: key, numeric (PK)
  - 23.2 deployment\_location\_id: numeric (FK)
  - 23.3 start\_time: timestamp
  - 23.4 end\_time: timestamp
- 24. **SupportRequest:** Weak
  - 24.1 support\_request\_id: key, numeric (PK)
  - 24.2 robot\_id: numeric (FK)
  - 24.3 issue\_details: text
  - 24.4 time\_reported: timestamp
- 25. **AIModel:** Strong
  - 25.1 aimodel\_id: key, numeric (PK)
  - 25.2 model\_name: alphanumeric
  - 25.3 service\_type: alphanumeric
  - 25.4 version\_label: alphanumeric
- 26. **AIRequest:** Weak
  - 26.1 ai\_request\_id: key, numeric (PK)
  - 26.2 aimodel\_id: numeric (FK)
  - 26.3 robot\_id: numeric (FK)
  - 26.4 requested\_at: timestamp
- 27. **AIResponse:** Weak
  - 27.1 ai\_response\_id: key, numeric (PK)
  - 27.2 ai\_request\_id: numeric (FK)

- 27.4 task\_id: numeric (FK)
- 27.5 role\_id: numeric(FK)
  
- 27.6 response\_payload: text
- 27.7 generated\_at: timestamp
  
- 28. **Hospital:** Weak (Subclass)
  - 28.1 hospital\_id: key, numeric (PK)
  - 28.2 deployment\_location\_id: numeric (FK, unique)
- 29. **Warehouse:** Weak (Subclass)
  - 29.1 warehouse\_id: key, numeric (PK)
  - 29.2 deployment\_location\_id: numeric (FK, unique)
- 30. **Home:** Weak (Subclass)
  - 30.1 home\_id: key, numeric (PK)
  - 30.2 deployment\_location\_id: numeric (FK, unique)
- 31. **Factory:** Weak (Subclass)
  - 31.1 factory\_id: key, numeric (PK)
  - 31.2 deployment\_location\_id: numeric (FK, unique)
- 32. **Department:** Weak
  - 32.1 department\_id: key, numeric (PK)
  - 32.2 hospital\_id: numeric (FK)
  - 32.3 department\_name: alphanumeric
- 33. **EmotionalPattern:** Weak
  - 33.1 emotional\_pattern\_id: key, numeric (PK)
  - 33.2 robot\_id: numeric (FK)
  - 33.3 ai\_model\_id: numeric (FK)
  - 33.4 pattern\_summary: text
- 34. **MaintenanceRecord:** Weak
  - 34.1 maintenance\_record\_id: key, numeric (PK)
  - 34.2 robot\_id: numeric (FK)
  - 34.3 technician\_id: numeric (FK)

- 34.4 RecordedAt: timestamp
- 34.5 Description: text
- 35. **Technician**: Strong
  - 35.1 technician\_id: key, numeric (PK)
  - 35.2 deployment\_location\_id (FK)
- 36. TechnicianName: Weak
  - 36.1. TechnicianNameID(PK):key,numeric
  - 36.2. FirstName :Text
  - 36.3. LastName:Text
  - 36.4. TechnicianID(FK)
- 37. **User\_Deployment\_Location**: Associative
  - 36.1 UserDeploymentLocationID: key, numeric (PK)
  - 36.1 deployment\_location\_id: numeric(FK)
  - 36.2 user\_id: numeric(FK)
- 38. **EmotionalAnalyze**: Associative
  - 37.1 EmotionalAnalyzeID: key, numeric (PK)
  - 37.2 emotional\_id: numeric(FK)
  - 37.3 ai\_request\_id: numeric(FK)
- 39. **PerformanceDiagnostic**: Associative
  - 39.1 PerformanceDiagnosticID: key, numeric (PK)
  - 39.2 diagnostic\_report\_id: numeric(FK)
  - 39.3 performance\_metric\_id: numeric(FK)
- 40. **EmotionalInteractionSession**: Associative
  - 40.1 EmotionalInteractionSessionID: key, numeric (PK)
  - 40.2 emotional\_id: numeric(FK)
  - 40.3 interaction\_id: numeric(FK)
- 41. **Technician\_Support\_Request**: Associative
  - 41.1 TechnicianSupportRequestID: key, numeric (PK)
  - 41.2 support\_request\_id: numeric(FK)
  - 41.3 technician\_id: numeric(FK)

- 42. **RobotChargingStation**: Associative
    - 42.1 RobotChargingStationID: key, numeric (PK)
    - 42.2 chargin\_station\_id: numeric(FK)
    - 42.3 robot\_id: numeric(FK)
  - 43. **logSession**: Associative
    - 43.1 logSessionID: key, numeric (PK)
    - 43.2 charging\_session\_id: numeric(FK)
    - 43.3 log\_id: numeric(FK)
  - 44. **HardwareMaintenance**: Associative
    - 44.1 HardwareMaintenancelD: key, numeric (PK)
    - 44.2 hardware\_id (FK)
    - 44.3 maintainence\_record\_id (FK)
  - 45. **ChargingStationSession**: Associative
    - 45.1 ChargingStationSessionID: key, numeric (PK)
    - 45.2 charging\_session\_id: numeric(FK)
    - 45.3 charging\_station\_id: numeric(FK)
  - 46. **PatientEmotionalRecord**: Associative
    - 46.1 PatientEmotionalRecordID: key, numeric (PK)
    - 46.2 emotional\_id: numeric(FK)
    - 46.3 hospital\_id: numeric(FK)
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## EER Diagram(Normalize)



## Database Schema Diagram

